

Predicting Skin Cancer Status

By: Hartejh Surikapuram, Anika Chowdhury, Lucy Borkowski,
Justine Shu, Sriya Donepudi
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Abstract

Skin cancer is the most common type of cancer in the U.S., and early detection can save the lives of many individuals. In this paper, we will discuss several statistical learning models to predict the type of skin cancer individuals have, to classify them as Benign or Malignant. We use a training dataset containing 50,000 observations and 49 predictors and a testing dataset of 20,000 observations and 49 predictors. Demographic, environmental, and lesion information are included as predictors in both datasets. We addressed missing data using median and mode, and then created Random Forest, Gradient Boosted Trees (GBM), and regularized logistic regression models. The performance of our models were tested in the Kaggle competition to find which of our models had the best accuracy. Our final model that we chose was a regularized logistic regression model with the highest accuracy of 60.65% on Kaggle, with our rank being 10th place, as of the end of the competition. While our final model achieved a moderate accuracy, we see that there is bias from our data cleaning and did not recognize the complexities in the relationships in the dataset, so there is a need for further research in clinical features and image screening for better predictions.

Introduction

Skin cancer is the most common cancer in the United States, affecting millions of people each year. Data shows that one in five Americans will develop skin cancer by the age of 70, and over two people die from the disease every hour in the U.S (American Academy of Dermatology, 2025; Skin Cancer Foundation, 2025). The survival rate for melanoma is very high when detected early, but assessing a person's risk can be difficult to do. Skin cancer screening is done through things like visual examination, dermoscopy, and biopsy, with diagnostic accuracy ranging widely depending on the method (Rigel et al., 2010). Some methods like biopsies are highly accurate, but they are more invasive and expensive, while less intrusive and costly methods like visual screenings yield lower diagnostic accuracy rates. Therefore, machine learning and model development could help improve this evaluation process, bettering the analysis of risk factors and identifying patterns. By improving early detection, there is potential to help save many lives.

In our project, we used a skin cancer dataset made of data describing a wide variety of variables, like skin tone and lesion size, that help to diagnose skin cancer. The dataset includes numerical and categorical variables relating to skin cancer, which help to further our models understanding of variable relationships and improve classification techniques. Our end goal was to develop a model that actually predicts whether a given data point is benign or malignant.

Data

The data is from Kaggle, from a skin cancer classification competition; it includes a large number of observations with many different types of predictors - demographic, environmental,

behavioral, etc. The training dataset has 50,000 rows and 50 variables whereas the testing data set has 20,000 rows and all the predictors without the Cancer label.

Before modeling, we explored the data's structure to understand the variable types and to see whether any of the variables needed special handling. We observed that the Cancer variable was roughly balanced (Benign $\sim 47.7\%$ and Malignant $\sim 52.3\%$); baseline accuracy was 0.5226 and baseline error rate was 0.4774. There were both numerical (18 numeric) and categorical (31 categorical) types, as well as variables that were numerical but actually represented categories. For example, the variable outdoor job is coded as 0 or 1 but reflects a Yes/No category, and the last digit of zipcode takes values from 0 to 9 but does not represent a meaningful numerical scale. Recognizing these distinctions was important during data preprocessing to ensure variables were treated appropriately in the modeling process. Missingness was evenly spread across predictors, and no predictor had more than 10 percent missing values, so we didn't drop any predictors due to excessive missingness and addressed missing values during later modeling steps.

Methods/Models

The training and testing skin cancer datasets contained missing values, the training had 196,042 missing values, and the testing had 78,256 missing values. From this we saw that the missing values were evenly distributed, so we did not want to delete variables completely so we tested a couple methods to address the missing values. We attempted to implement missForest, MICE, and median for replacing numerical variables and mode for categorical variables. While using each method we applied them in our models, and through the comparisons, we found that median and mode led to higher accuracy for our models while being efficient as well, so we ultimately decided to continue using median and mode for all our models. With our two data sets and missing values addressed, we created several models, including Random Forest, Gradient Boosting, and logistic regression models to find the model with the best accuracy in classifying the skin cancer type.

1. Random Forest

The given dataset is quite large, so we wanted a model that would be able to handle complex relationships with numerical and categorical predictors, and also allow us to perform feature selection. From this, we tried a Random Forest model using all 49 predictors in the training dataset and used 500 trees. For our full Random Forest model, we plotted the MCR plot, shown in Figure 1, which shows that the errors for both skin cancer types stabilized by 100 trees, so there was convergence, and the OOB error stabilized at around 0.45. We also see that before the errors stabilized, there is a difference in the accuracy of the model classifying the two classes, as the Malignant error decreased as more trees were added, while the Benign error increased. So we created a confusion matrix, Figure 2, where we can see that Malignant was classified with a lower

error rate of 33.12% while Benign has an error rate of 52.04%. From this, we see that there is a bias for Malignant in our full model's classification. The full model had moderate accuracy in classifying the skin cancer type, with a training accuracy of 60.76% and a Kaggle testing accuracy of 59.84%.

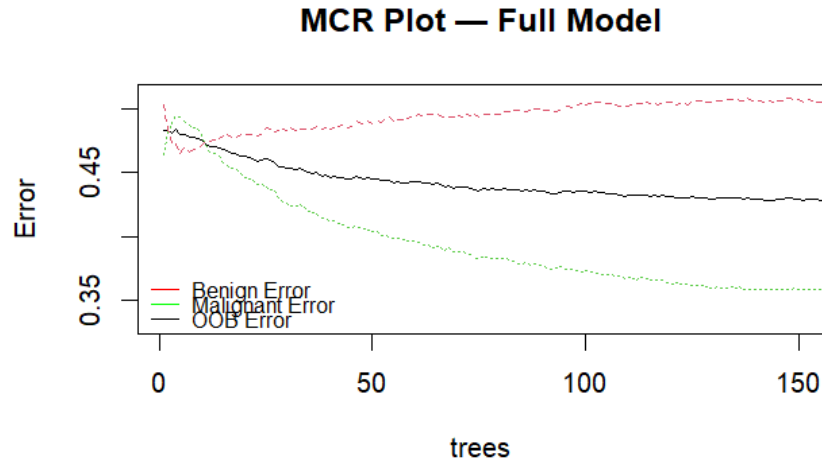


Figure 1 - Misclassification Rate (MCR) Plot for the Full Random Forest Model

Full Model Training Confusion Matrix			
	Benign	Malignant	Class Error
Benign	11684	12184	0.5104743
Malignant	8656	17476	0.3312414

Figure 2 - Confusion Matrix for the Full Random Forest Model

Considering the full model accuracy, we wanted to use feature selection with variable importance to see if it could improve our classification accuracy. Looking at the variable importance plot, Figures 3 and 4, we see that demographic factors and clinical information are ranked higher for classification accuracy, while geographical predictors ranked higher for separation power. We then decided to create a Random Forest model using the top 15 predictors using both MeanDecreaseAccuracy and MeanDecreasesGini for a reduced model. This way, we can address potential overfitting by removing predictors that were adding noise. For our 15 predictor Random Forest model, we looked at its MCR model, Figure 5, in which we see a similar pattern in the errors as the full model, as the trees increase, Benign error continues to increase, and Malignant decreases, but slightly less than the full model. However, from the confusion matrix of the 15 predictor model, Figure 6, Benign error is 55% and Malignant error is 35%, so this reduced model is weaker in classifying both classes than the full Random Forest

model. For this model, our training accuracy was 60.02% and our Kaggle testing accuracy was 56.005%. Neither of the Random Forest models gave an accuracy we were satisfied with, so we continued to explore other models to find one that may better address the complexities of the data set.

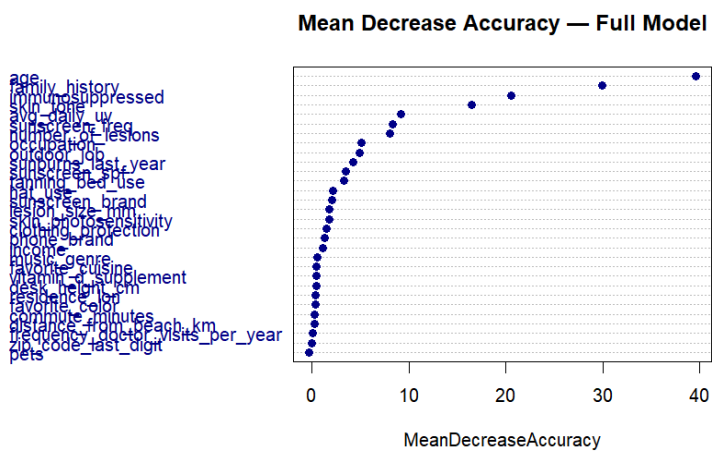


Figure 3 - Mean Decrease Accuracy Variable Importance

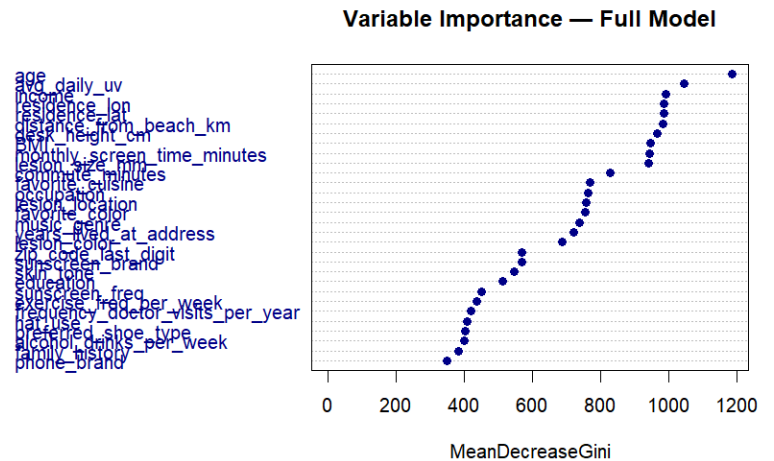


Figure 4 - Mean Decrease Gini Variable Importance

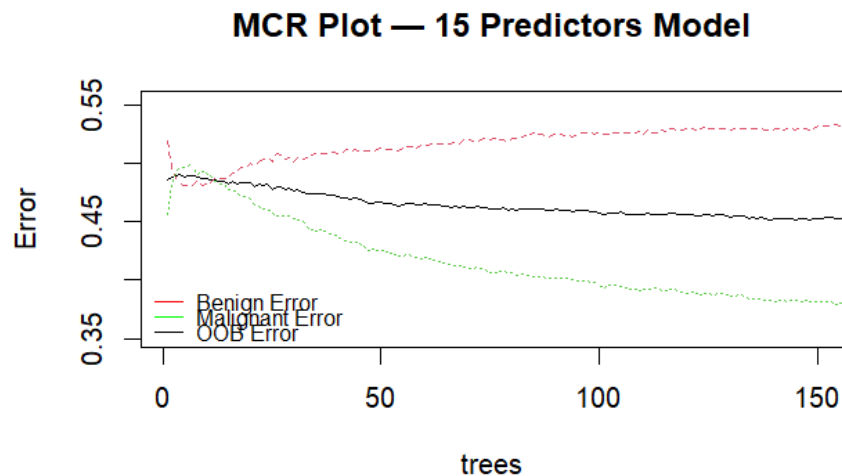


Figure 5 - Misclassification Rate (MCR) Plot for 15 Predictor Random Forest Model

15 Predictor Model Training Confusion Matrix			
	Benign	Malignant	Class Error
Benign	10686	13182	0.5522876
Malignant	9161	16971	0.3505664

Figure 6 - Confusion Matrix for 15 Predictor Random Forest Model

2. Gradient Boosting Machine (GBM)

The dataset is very complex – there are many weak predictive variables and potentially also nonlinear interactions – so we next explored Gradient Boosting Machines (GBM) as another tree-based model. Compared to bagging and random forests, models that fit a lot of trees independently on bootstrap samples, gbm involves building trees sequentially. Each new tree works on correcting the errors of the previous ones. This slow, iterative process of learning helps GBM models capture subtle nonlinear relationships that single trees can sometimes miss.

Before fitting the model, we handled missing values consistently across both the training and testing datasets by doing median imputation for all the numerical predictors and mode imputation for all the categorical variables. We also removed the ID variable, since it does not have predictive information, recoding the response as a binary variable (0 = Benign, 1 = Malignant) so that it matches the Bernoulli loss in boosting.

We tuned several hyperparameters that govern how a GBM learns:

- **Number of trees** (n.trees): the number of boosting iterations – larger values give the model a greater number of opportunities to decrease error
- **Interaction depth** (interaction.depth): controls tree size – higher-depth trees allow for higher-order interactions
- **Shrinkage** (shrinkage): the learning rate – smaller values (slowed updating) often lead to more stable models
- **Minimum observations in terminal nodes** (n.minobsinnode): helps prevent overly small, high-variance leaf nodes
- **Cross-Validation folds** (cv.folds): determines number of boosting iterations model actually uses

We experimented with many combinations of those hyperparameters. We noticed that increasing the number of trees initially decreased the training deviance but it eventually caused the cross-validated deviance to increase, which implies overfitting. After multiple attempts of changing parameters, our best-performing model used 8,000 trees, depth = 5, shrinkage = 0.01, and 10-fold CV. The cross-validated deviance plot (Figure 7) shows that although the model was allowed to go up to 8,000 trees, the optimal stopping point was actually a lot earlier – around 1,400 - 1,600 iterations of boosting. With boosting, setting a big maximum number of trees seems to give the CV process flexibility in choosing the best iteration without limiting the model too early.

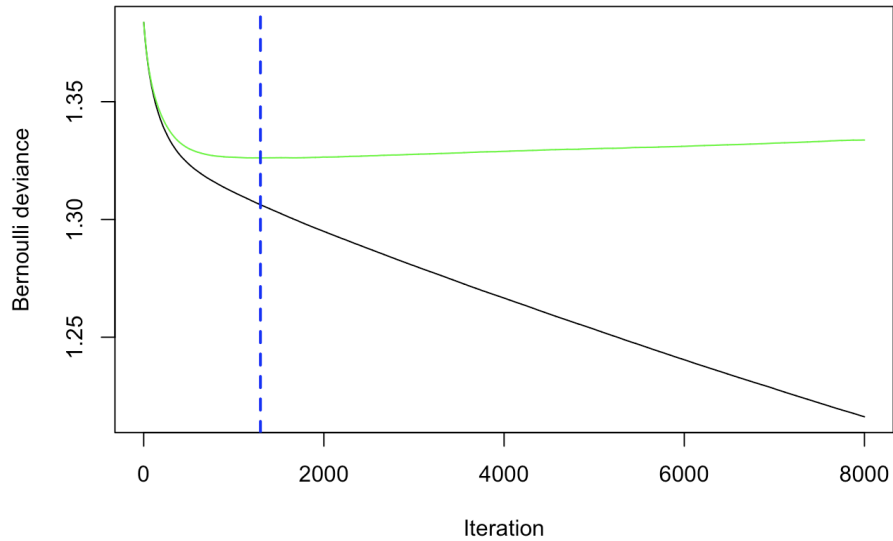


Figure 7: Cross-Validated Deviance Curve for GBM model

Figure 7 shows the training Bernoulli deviance (black line) and the cross-validated deviance (green line) along the number of boosting iterations (x-axis). The vertical, dotted blue line represents the optimal number of iterations, and thus trees, that 10-fold CV chooses. The model was allowed to go to a large amount (8,000) trees before choosing the stopping point to prevent overfitting.

With the selected optimal CV iteration, we produced predictions for the Kaggle testing dataset. This boosted model got a higher test accuracy than all the other gbm models we tried and the full and reduced Random Forest models; it had a Kaggle score of 60.415%. These results suggest that GBM, compared to RandomForest and a few other models, was better able to extract the predictive signal from the weak predictors in the dataset, although overall accuracy could still be improved.

3. Logistic Regression

For the logistic regression part of the project, our goal was to build a strong and interpretable baseline model and then improve it step by step through experimentation and feature selection. We started broadly and gradually narrowed the model based on performance and statistical significance.

We first fit a full logistic regression model using all available predictors. As a baseline classifier, this full logistic model actually performed relatively well compared to other approaches we tested, including Random Forest and Gradient Boosting. However, while it was competitive, the accuracy was still far from ideal.

After that, we experimented with several variations of logistic regression to try to improve performance and reduce noise in the model. These included Lasso, stepwise selection, and subset selection. The goal of these methods was to control overfitting and identify which predictors were contributing meaningfully versus adding randomness.

After testing these approaches, we found that a simpler logistic regression model using only statistically significant predictors from the full model performed surprisingly well.

This version of the model used 18 statistically significant predictors, including:

- Demographic factors such as age and skin tone
- UV exposure and sun related behaviors like sunscreen/tanning bed use
- Protective behaviors such as wearing hats and protective clothing
- Medical and lesion characteristics like lesion size, number of lesions, and skin photosensitivity

Training Confusion Matrix			
	Benign	Malignant	Class Error
Benign	13006	9174	0.4136
Malignant	10862	16958	0.3904

Figure 8 – Confusion Matrix for 18 Predictor Logistic Regression Model

As shown in Figure 1, the logistic regression model has similar class error rates for both classes, with benign cases having a slightly higher error (41.4%) than malignant cases (39.0%). This indicates that misclassifications are fairly evenly distributed, though the model is marginally better at identifying malignant cases. Additionally, this model achieved a training accuracy of about 59.9% and a Kaggle test accuracy of 60.545%, making it one of the strongest performing models we had at that stage of the project.

Final Model

Among the models that were tried and tested, the final candidate model is the **ridge regularized logistic regression model** due to its superior performance. Using the dataset, a variant of the logistic regression model managed to yield a 0.6065 accuracy rate on Kaggle.

To improve model performance, we applied extensive feature engineering. The original dataset contained 49 predictors, which was expanded to 57 predictors after introducing engineered variables designed to better capture skin cancer risk mechanisms. The engineered features include:

- Adjusted UV exposure, which accounts for the protective effects of frequent sunscreen use by reducing effective UV exposure for individuals who reported regular sunscreen use.
- Cumulative outdoor exposure, combining average daily UV levels, outdoor job status, and occupation type to reflect long-term environmental exposure.
- Skin vulnerability indicators, identifying high-risk individuals with fair or light skin tones who reported recent sunburns.
- Age groupings, allowing for non-linear effects of age on cancer risk rather than assuming a strictly linear relationship.
- Protective behavior scores, summarizing sunscreen use, hat use, and protective clothing into a single measure of sun protection.
- Lifestyle risk measures, incorporating smoking status, alcohol consumption, and immunosuppression to reflect broader health-related risk factors.
- Interaction effects, capturing how UV exposure varies by skin tone and how outdoor exposure differs across age groups.

After feature engineering, all predictors were standardized and used to fit a ridge-regularized logistic regression model. Ridge regression was chosen because it helps stabilize coefficient estimates without removing variables entirely. The regularization parameter lambda was selected using cross-validation. Several lambda values were evaluated, and the value corresponding to the minimum cross-validated error performed best at 0.006768, which consistently outperformed alternative choices.

Training Confusion Matrix			
	Benign	Malignant	Class Error
Benign	12968	9077	0.4117
Malignant	10900	17055	0.3899

Figure 9 – Confusion Matrix for Final Ridge Logistic Regression Model

Using this lambda value, the final ridge logistic regression model achieved a training accuracy of approximately 60.0 percent. The confusion matrix showed balanced classification performance across outcome classes, with a Benign class error rate of 41.17 percent and a Malignant class error rate of 38.99 percent. This indicates that misclassifications were fairly evenly distributed, with the model slightly better at identifying malignant cases.

While methods like gradient boosting and random forest are capable of capturing non-linear relationships to a more efficient degree, their performance did not exceed logistic regression's. This might be due to the relationship between the predictors and malignancy status being sufficiently captured by linear assumptions.

Data Analysis

The final model identified several significant predictors that are aligned with the presence of skin cancer malignancy. These predictors span demographic/environmental, behavioral, medical, and lesion-specific domains. Demographic and environmental factors such as age, urban versus rural, years lived at current address, and skin tone were significant. This reflects current known associations about sun exposure and environmental context. Behavioral factors such as hat use, clothing protection, tanning bed use, and sun burns in the past year shows the role of lifestyle choices and preventative measures affecting melanoma risks.

Discussion

Although the final logistic regression model outperformed the other models, its accuracy is still unfortunately insufficient for clinical usage. In real-world applications, diagnostic tools tend to exceed 90% accuracy in order to minimize the number of false negatives during diagnosis. A model that is only slightly better than a coin flip will lead to delayed life-saving treatment. Future work should incorporate richer clinical features as well as medically informed variables into consideration. While logistic regression provided valuable insight into the complexities of malignancy classification, this current approach should be treated as exploratory rather than diagnostic.

Limitations

It is important to consider limitations to the accuracy and interpretability of our models and results. Firstly, our models relied on median and mode imputation to handle missing data, a method that can oversimplify datasets. By replacing missing values with central tendencies, important variation may have been lost and bias may have been introduced, and while our best model did use engineered features intended to capture interactions and non-linear patterns, their usefulness is also limited by the original variables and imputations. Additionally, it is possible the engineered features introduced additional noise and bias. Furthermore, when we tried strategies to combat potential simplicity, like MICE and missForest, it did not lead to better model performance. Secondly, computational limitations narrowed how extensively we were able to tune our Random Forest and Gradient Boosting models. These models required significant time to run due to the size of the dataset and the number of predictors, which limited our ability to explore a wide range of hyper parameters that could have improved model performance. Furthermore, even with tuning we did, both our Random Forest and GBM models showed signs of overfitting. Moreover, the lack of interpretability these models present is yet another limitation to using them. While they can capture complex, non-linear relationships, they are difficult to interpret, making it challenging to evaluate the real world usefulness of their predictions. On the other hand, logistic regression is more interpretable, but it is poor at modeling non-linear relationships and is sensitive to multicollinearity, which may have affected

our various models' performance. Thirdly, model performance varied depending on things like the specific train and test split and the random seed used, impacting model and result reproducibility. This suggests that the dataset may not consistently produce reliable classifications. Finally, some predictors that logically seemed important were statistically insignificant in our models, suggesting another potential model limitation.

Conclusion

In conclusion, our analysis explored several modeling and tuning approaches to predict skin cancer status, including Random Forests, Gradient Boosting, and multiple forms of logistic regression. While an initial logistic regression model using only statistically significant predictors achieved a Kaggle accuracy of approximately 60.545%, we were able to further improve performance to 60.650% by implementing a ridge logistic regression model and feature engineering, reaching 10th place on Kaggle. This represents a meaningful improvement over our baseline accuracy of 52.26%, yet the overall predictive performance still remains low for actual use in the real world. More complex models such as Random Forest and GBM produced lower accuracy results while also being much more computationally expensive, making ridge logistic regression the most practical, interpretable, and best overall choice. The moderate accuracy across our models suggests that the given data set does not fully capture the complex biological, behavioral, and environmental factors that influence skin cancer risk in full. As a result, our final model should be viewed as a 'baseline model' that still needs further improvement. Future improvements would likely require richer data, like image-based data, combined with more advanced imputation strategies we can't implement.

Overall, this project highlights both the potential and limitations of using data and traditional machine learning methods for medical predictions, emphasizing the importance of high quality and relevant variables in achieving meaningful and accurate results.

References

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